

Bi-Level Stochastic Optimization in Distribution Network Planning: Decision-Dependent Distributionally Robust Joint Chance Constrained for Uncertainty Handling

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Abstract—Heterogeneous distribution energy resources, such as renewable energy sources and electric vehicles (EVs) impose prominent challenges to active distribution network planning. This paper proposes a bi-level stochastic optimization framework that enables the coordinated optimization of renewable generation, flexible loads, grid capacity, and energy storage through an EV integration mechanism with rejection sets. A mixed-integer non-linear programming model is constructed to minimize total planning costs under technical and operational constraints, where upper-level investment decisions are coupled with lower-level operations. For the long-short term uncertainties in EV demand and renewable energy output, a scenario-based decision-dependent distributionally robust joint chance-constrained approach is employed for effective characterization. Finally, the proposed bi-level model is transformed into a single-level optimization problem via linearization, significantly reducing computational complexity. Simulation results on the 25-node system, IEEE 141 and IEEE 300 system demonstrate that the proposed approach significantly reduces the total planning cost and alleviates voltage violations and line congestion, while maintaining computational scalability.

Index Terms—distribution network planning, electric vehicle integration, uncertainties, decision-dependent, distributionally robust joint chance constrained.

I. INTRODUCTION

High penetration of distributed energy resources (DERs) transforms distribution networks into stochastic, bidirectional systems, challenging deterministic planning and increasing operational risks. Virtual power plants (VPPs) enable coordinated flexibility by aggregating DERs, making their integration crucial for resilient and cost-effective active distribution network planning [1].

A diverse array of resources are investigated for systematic ADN planning in recent literature. Firstly, for distributed generators (DGs): Li *et. al* [2] address energy equity by proposing siting and sizing methods for DGs. An efficient planning approach is introduced based on the 2m+1 Point Estimate Method, integrating loss sensitivity and voltage stability to determine optimal DG locations [3]. Secondly, for energy storage system (ESS): A two-stage stochastic bilevel model that co-optimizes the siting and sizing of ESS (first stage) with operational arbitrage (second stage) is presented. The operational stage employs a bilevel structure, where the upper

level maximizes arbitrage profits and the lower level clears the market [4]. ADMM algorithm is utilized to coordinate planning between microgrid cluster operators and shared energy storage operators [5]. Thirdly, concerning grid line investment: long-term infrastructure investments are integrated with short-term operational strategies in multi-stage planning [6]-[8]. Lastly, for loads like EVs, a two-stage stochastic planning framework incorporating vehicle to grid (V2G) characteristics is proposed, using scenario-based methods to address EV stochasticity and multi-energy uncertainties, extending EV charging station (EVCS) planning to electricity-heat coupled multi-energy systems [9]. An EV diffusion function is developed in [10], employing a decision-dependent ambiguity set to quantify the bidirectional influence of FCS layout on EV adoption rates. However, existing studies rarely consider the joint planning modeling of renewable energy sources (RES), EVCS, distribution lines, and ESS, neglecting the interdependencies of heterogeneous sources.

At the operational layer, the development of VPP has evolved from basic distributed energy aggregation to sophisticated radial ADN systems capable of bidirectional grid services and market participation. An orderly charging model for EV-aggregated VPP scheduling is proposed in [11]. Under the novel power system paradigm, data-driven planning methods for hybrid energy storage systems in high-penetration renewable energy environments are introduced, aiming to mitigate the uncertainty of RES [12], [13]. With the increasing maturity of EV-based VPP scheduling studies, a charging pile planning model accounting for human-factor-influenced V2G participation in multi-energy systems is developed [9]. However, few studies have explored aggregated EV VPPs' application in ADN planning scenarios, nor have they leveraged idle EV resources to enhance planning flexibility (aggregated EVs) for the optimal planning. Meanwhile, during routine operation of EVCS after planning stage, voltage/branch flow limit violation issues are neglected in existing studies.

Problems involving uncertain parameters are typically addressed through stochastic programming (SP), robust optimization (RO) and distributionally robust optimization (DRO). SP requires the probability distributions of uncertain parameters. The effective characterization of uncertainty persists as a foundational challenge in contemporary power system planning [4], [9]. RO usually obtains the result under the worst scenario within an uncertainty set, prone to falling into a conservative dilemma [10], [12]. Therefore, stochastic distributionally robust joint chance-constrained (DRJCC) model is proposed [14]. These approaches depict the uncertainty with an ambiguity set, a family of probability distributions, to

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balance the economy and robustness with satisfactory computational efficiency. However, most existing DRJCC-based studies apply the ambiguity set to exogenous uncertainties, such as renewable generation errors, load fluctuations, or pre-existing distributed resources, and the uncertainty set is usually constructed independently of planning decisions. In contrast, in ADN planning with new WT/PV installations, the magnitude and distributional characteristics of renewable forecast errors are directly affected by the installed capacities. Therefore, the uncertainty set should be endogenously linked with planning decisions rather than treated as a fixed model input.

In addition to the aforementioned points, existing studies mainly focus on either short term or long-term uncertainties [16]–[18], neglecting the coupling of long-term and short-term uncertainties. Wind turbine (WT) fluctuation uncertainty is addressed, primarily focusing on short-term variability of RES [15], [16]. A stochastic power flow model incorporating Gram-Charlier expansion is employed to handle short-term power flow fluctuations [17]. SP under complete distribution knowledge is adopted, considering only long-term multi-scenario uncertainty while ignoring short-term fluctuations [18].

Finally, bilevel optimization has been widely used to capture the interactions among different stakeholders in power system planning and operation [19]–[22]. However, existing bilevel planning studies mainly focus on market-clearing mechanisms, traffic–power coupling [23], or single-resource investment decisions [2]–[9], while the coordinated planning of RES, ESS, EVCS, and grid reinforcement under decision-dependent distributional ambiguity remains insufficiently explored.

TABLE I
COMPARISON BETWEEN THE PROPOSED METHOD AND EXISTING STUDIES

Ref.	System asset considered	Uncertainty model	EV fleet	Solution
[2]	DG	SP	-	Bi-level→MILP
[4]	ESS	SP	-	Bi-level→MILP
[6]	DG+Grid+Sub	Scenario	-	MILP
[9]	EVCS	SP	✓	MINLP
[10]	EVCS+RES	Adaptive DRO	✓	MILP/Benders
[12]	ESS+RES	RO	-	MILP/ICCG
[14]	DG+HSS+RES	DRJCC+Scenario	-	MISOCP
[15]	DG+Grid+Sub	DRCC	-	MILP
[19]	Grid+RES+Sub	Scenario	-	MILP
[23]	Grid+EV+PV	Scenario	✓	Big-M/ADMM
[25]	5G BS+RES	Interval	-	Heuristic
[38]	RES+Data center	DRJCC	-	MISOCP
This paper	RES+ESS+EVCS+Grid	DD-DRJCC+Scenario	✓	Bi-level+bigM→MILP

Note: Substations(Sub), hydrogen storage system(HSS) 5G BS(5G Base Station)

This paper bridges these research gaps by developing a bilevel ADN planning model that actively integrates EV-aggregated VPP participation and considers long-short term uncertainties. Bilevel optimization is then reformulated into single level problem, which is a computationally tractable mixed-integer linear programming (MILP) problem solvable via commercial solvers. The key contributions of this paper

include:

1) A unified bi-level planning model is proposed to simultaneously co-optimize the placement and sizing of renewable generators, ESS, EVCS, and grid reinforcements. The framework shifts from traditional single-resource planning in [3], [4], [6], [10] to a holistic infrastructure design that fully accounts for heterogeneous assets' joint optimization.

2) An incorporation of EV-aggregated VPPs is introduced into the planning framework, capturing spatial-temporal flexibility through a rejection-priority mechanism. Compared to Ref. [10]–[12], the proposed approach enables dynamic charging and discharging coordination to mitigate voltage and power flow violation, and enhances the cost-effectiveness of DER deployment.

3) A long–short-term hybrid uncertainty characterization approach is developed. Distinct from existing SP, RO, and DRJCC-based approaches [2], [14]–[17], long-term variations are represented by scenario-reduced typical days, while short-term renewable fluctuations are handled through a decision-dependent DRJCC framework. In this formulation, the ambiguity set is endogenously linked with planned RES capacities, enabling the model to capture capacity-dependent renewable uncertainty and improve the balance between planning economy and operational reliability.

The rest of the paper is organized as follows. Section II describes the model of aggregated EVs and uncertainty modeling. Section III presents a stochastic bilevel optimization model. Section IV proposes transformation of bilinear constraints and bilevel problems. Simulation results and analyses are presented in Section V. Finally, conclusions are presented in Section VI.

II. MODEL PRELIMINARY

This section distills the essential physical and stochastic ingredients. We first span a single vehicle's arrival-departure-SoC dynamics up to its aggregated VPP feasible region, and construct an ambiguity set with scenario reduced RES profiles.

A. Modeling of EV users

The arrival-departure times and state-of-charge (SoC) levels of EVs are inherently user-determined stochastic processes. Following the methodologies established in [9]–[10], Gaussian normal distributions are adopted to model the uncertainty in EV charging demand.

$$y_n = \mathcal{N}(\mu_n, \sigma_n^2), \forall y_n \in \{t_{ari}, SoC_{ari}\} \quad (1)$$

$$Y_m = \mathcal{N}(\mu_m, \sigma_m^2 | y_n), \forall Y_m \in \{t_{dep}, SoC_{dep}\}$$

where y_n denotes the random variables including the arrival time and arrival SOC(t_{ari} and SoC_{ari} , respectively), and Y_m denotes the departure's part. Then, because of the number of EV at time slot t may be smaller than the number of charging piles, a rejection set $\mathcal{R}$ is introduced as a critical filtering mechanism to ensure the optimal resource allocation in EV charging system.

$$\mathcal{R} = \bigcup_{t \in T} \mathcal{R}_t \quad (2)$$

$$\mathcal{R}_t = \left\{ (u,i) \in \mathcal{A}_t \mid \begin{array}{l} t = t_{u,i,ari} \wedge \\ \sum_{(v,j) \in \mathcal{A}_t} \mathbb{1}_{[t_{v,j,ari}, t_{v,j,dep}]}(t) \geq n_i^{EVCS} \wedge \\ \forall (v,j) \in \mathcal{A}_t \setminus \mathcal{R}_t, \\ prio(v,j) > prio(u,i) \end{array} \right\} \quad (3)$$

where $\mathcal{R}_t$ is the time-slot rejection subset, $\mathcal{A}_t$ is the set of EVs arriving at time slot t , $\mathbb{1}_{[a,b]}(t)$ is indicator function, n_i^{EVCS} is the number of EV charging piles. when $t \in [a,b]$ returns 1, else 0. And the priority function $prio(u,i)$ is given by:

$$prio(u,i) = \frac{E_{s,u,i,t,ari} - E_{s,u,i,t,dep}}{\eta_{u,i}^{ch} \cdot (t_{u,i,dep} - t_{u,i,ari})}. \quad (4)$$

Eq.(3) defines EVs (u,i) are rejected by EVCS when three conditions are satisfied: 1. EV's arrival time $t_{u,i,ari}$ should precisely equal to current scheduling time interval t , expressed as $t=t_{u,i,ari}$. 2. Rejection is triggered only when the number of arrived EVs exceed station's capacity n_i^{EVCS} . 3. The vehicle (u,i) is rejected if it exhibits the lowest priority score within the arrival set $\mathcal{A}_t$, as quantified by metric $prio(u,i)$ in (3), meaning that, for all vehicles (v,j) that are not rejected in the current time period t , their priority is higher than that of the rejected vehicle (u,i) .

The rejection set can be interpreted as an admission-control mechanism under limited EVCS capacity. For example, consider an EVCS with two charging piles. EV 1 arrives at 8:00 with an SoC requirement from 10% to 60% and departs at 14:00; since the station is not full, it is directly admitted. At 9:00, EV 2 and EV 3 arrive simultaneously, while only one charging pile remains available. EV 2 departs at 18:00 with an SoC requirement from 20% to 80%, whereas EV 3 departs at 13:00 with an SoC requirement from 30% to 70%. The EVCS then admits one of them according to the priority function and places the other in the rejection set. For instance, if charging urgency is emphasized, EV 3 may be prioritized due to its earlier departure time; if VPP flexibility is emphasized, EV 2 may be prioritized due to its longer parking duration. Priority function can be customized according to different requirements.

This mechanism also links charging-piles' planning with the aggregated EV feasible region: excessive piles may cause unnecessary investment, while insufficient piles may exclude many EVs from aggregation, limiting charging service and VPP flexibility. Consequently, both over- and under-investment can increase the total planning and operation cost, as further verified by the sensitivity analysis on charging-pile numbers.

For each EV, the charging and discharging actions must lie within its individual feasible energy region, which is determined by its arrival time, departure time, state-of-charge limits, and charging/discharging power constraints.

$$E_{s,u,i,t} = \begin{cases} 0, & t < t_{u,i,ari}, t > t_{u,i,dep} \\ E_{s,u,i,t-1} + P_{s,u,i,t}^{g2v} \eta_{u,i}^{ch} \Delta t \\ - P_{s,u,i,t}^{v2g} \Delta t / \eta_{u,i}^{dch}, & \text{otherwise} \end{cases} \quad (5)$$

$$\bar{E}_{s,u,i,t} = \begin{cases} 0, & t < t_{u,i,ari}, t > t_{u,i,dep} \\ \min\{E_{s,u,i,t,ari} + \eta_{u,i}^{ch} \bar{P}_{u,i}^{g2v} (t - t_{u,i,ari}) \\ , E_{s,u,i,t}^{\max}\}, & \text{otherwise} \end{cases} \quad (6)$$

$$\underline{E}_{s,u,i,t} = \begin{cases} 0, & t < t_{u,i,ari}, t > t_{u,i,dep} \\ \max\{E_{s,u,i,t}^{\min}, E_{s,u,i,t,ari} - \bar{P}_{u,i}^{v2g} (t - t_{u,i,ari}) / \eta_{u,i}^{dis} E_{u,i}^{desire} \\ - \eta_{u,i}^{ch} \bar{P}_{u,i}^{g2v} (t_{u,i,dep} - t)\}, & \text{otherwise} \end{cases} \quad (7)$$

$$\begin{cases} 0 \leq P_{s,u,i,t}^{g2v} \leq \bar{P}_{s,u,i,t}^{g2v}, 0 \leq P_{s,u,i,t}^{v2g} \leq \bar{P}_{s,u,i,t}^{v2g} \\ E_{s,u,i,t} \leq E_{s,u,i,t} \leq \bar{E}_{s,u,i,t}, P_{s,u,i,t}^{g2v} \cdot P_{s,u,i,t}^{v2g} = 0 \end{cases} \quad (8)$$

where (5) defines the energy state $E_{s,u,i,t}$ of EV u at node i during time step t . For t outside the arrival $t_{u,i,ari}$ and departure $t_{u,i,dep}$ times, the energy state is 0. (6) defines the maximum allowable

energy $\bar{E}_{s,u,i,t}$ for EV u at time t . (7) defines the minimum allowable energy $\underline{E}_{s,u,i,t}$ for EV u at time t . (8) regulates real-time power flow (G2V/V2G) and energy-based safety limits over time. They ensure feasible and efficient EV-grid integration in optimization models.

B. Scenario Reduction

Uncertainty is modeled on two coupled time scales. Long-term uncertainty is represented by annual scenarios of EV penetration and RES derived via scenario reduction from historical datasets.

The minimization objective function of the K-means algorithm is written as:

$$\min J = \sum_{k=1}^K \sum_{d=1}^{N_d} \|x_d - u_k\|^2, u_k = \frac{1}{N} \sum_{d=1}^{N_d} x_d \quad (9)$$

x_d is actually the statistical property of one day, and u_k is clustering center, which also represents the typical day. This approach preserves key statistical characteristics while reducing computational burden.

C. Joint Chance-Constrained Decision-Dependent Uncertainty

Short-term uncertainty represents intra-day renewable output fluctuations, modeled by moment-based ambiguity sets within each typical day. By nesting the short-term DRJCC inside the long-term scenario, the model captures both structural investment-level uncertainty and operational-level volatility. The specific characterization of the two levels of uncertainty is illustrated in Fig. 4.

The ambiguity set is defined as:

$$\mathcal{P} = \left\{ \mathfrak{P} \mid \begin{array}{l} \bar{\mathfrak{a}}[\xi_t] = \mu_t \\ \bar{\mathfrak{a}}[(\xi_t - \mu_t)(\xi_t - \mu_t)] = \Sigma_t \end{array} \right\} \quad (10)$$

where $[\xi_t]$ is the expectation on the probability distribution. The vector $\mu_t \in \mathbb{R}^{|\mathcal{W}| \times 1}$ is the mean and the matrix $\Sigma_t \in \mathbb{R}^{|\mathcal{W}| \times |\mathcal{W}|}$ is covariance and positive semidefinite.

Since WT-PV generation uncertainty depends on installed capacity, the error is modeled as a capacity-dependent random variable. For tractability, the ambiguity set is assumed independent of capacity, yielding a linear error formulation.

This assumption is supported by empirical evidence showing near-linear scaling between forecast errors and installed capacity [27], while remaining computationally efficient compared with nonlinear alternatives [28], [29].

Total error is modeled as follows:

$$\Omega_{s,t}(\mathbf{n}) = \mathbf{n} \tilde{\xi}_{s,t} = \begin{bmatrix} n_1, \dots, n_i \end{bmatrix} \begin{bmatrix} \xi_{s,1,t} \\ \vdots \\ \xi_{s,i,t} \end{bmatrix} \quad (11)$$

$$\tilde{P}_{s,i,t}^{DG} = \hat{P}_{s,i,t}^{DG} - \alpha_{s,i,t} \Omega_{s,t}(\mathbf{n}) \quad (12)$$

$$\tilde{P}_{s,i,t}^{WT} = \hat{P}_{s,i,t}^{WT} + n_i \xi_{s,i,t} \quad (13)$$

where in scenario s , the error term at node i is denoted as $\xi_{s,i,t}$, $\tilde{\xi} \in \mathbb{R}^{|w| \times 1}$ is a real-valued vector with dimension equal to the cardinality of set w (the collection of WTs). $-\alpha_{s,i,t} \Omega(\mathbf{n})$ is adjusted power for deviation mitigation, $\mathbf{n}$ is the renewable generators' planning variable vectors.

III. PROBLEM FORMULATION

The formulation of the objective covers capital expenditure and operation aspect while satisfying the technical constraints. Both maintenance cost and penalty cost for PV, WT curtailment are considered in the operation expenditure and several technical constraints are checked to ensure the feasibility of the proposed planning solution.

A. Objective Functions of Upper and Lower Levels

Upper-level: Total cost modeling in DSO planning:

$$\min C_1 = C^{inv} + C^{line} + C^{ope} + C^{aban} + C^{DG} - B^{EV} \quad (14)$$

$$C^{DN} = N_d \sum_{s \in S} \rho_s \sum_{i=1}^{N_{bus}} \sum_{t=1}^T P_{s,i,t,\tau}^{DN} \pi_t^{DN} \quad (15)$$

$$C^{O\&M} = N_d \sum_{s \in S} \rho_s \sum_{i=1}^{N_{bus}} \sum_{t=1}^T \tilde{P}_{s,i,t,\tau}^{RES} \pi_t^{RES} \quad (16)$$

$$C^{aban} = N_d \sum_{s \in S} \rho_s \sum_{i=1}^T \pi_a (P_{s,i,t,\tau}^{PVa} + P_{s,i,t,\tau}^{WTa}) \Delta t \quad (17)$$

$$C^{DG} = N_d \sum_{s \in S} \rho_s \sum_{i=1}^{N_{bus}} \sum_{t=1}^T (\pi_t^{DG-P} P_{s,i,t,\tau}^{DG} + \pi_t^{DG-Q} Q_{s,i,t,\tau}^{DG}) \quad (18)$$

$$B^{EV} = N_d \sum_{s \in S} \rho_s \sum_{i=1}^{N_{bus}} \sum_{u=1}^{N_{EV}} \sum_{t=1}^T (\pi_t^{dch} P_{s,u,i,t,\tau}^{v2g} - \pi_t^{ch} P_{s,u,i,t,\tau}^{g2v}) \quad (19)$$

where C^{DN} denotes procurement cost of electricity from upper grid in ADN, $C^{O\&M}$ denotes operation and maintenance costs of RES, C^{aban} is cost of curtailing wind and solar power, C^{DG} is the cost of traditional DG, B^{EV} denotes the operational revenue of EVCS under scenario s . N_d is the total number of days within the planning horizon, typically set as 365 days (equivalent to one calendar year). ρ_s signifies the probability associated with each scenario s .

Investment costs for DER planning by the DSO:

$$C^{line} = \sum_{\tau=0}^{T_p-1} (N_y - \tau N_p) PF(N_y - \tau N_p) \sum_{ij} c_{ij}^{line} (n_{ij,\tau+1}^{line} - n_{ij,\tau}^{line}) S_0^{line} \quad (20)$$

$$C^{inv} = \sum_{\tau=0}^{T_p-1} (N_y - \tau N_p) PF(N_y - \tau N_p) \sum_{i=1}^{N_{bus}} \left(c_i^{PV} (n_{i,\tau+1}^{PV} - n_{i,\tau}^{PV}) S_0^{PV} + c_i^{WT} (n_{i,\tau+1}^{WT} - n_{i,\tau}^{WT}) S_0^{WT} + c_i^{ESS} (n_{i,\tau+1}^{ESS} - n_{i,\tau}^{ESS}) S_0^{ESS} + (c_i^{inv} + c_i^{mnt}) x_i (n_{i,\tau+1}^{EVCS} - n_{i,\tau}^{EVCS}) \right) \quad (21)$$

where C^{line} denotes the line investment cost. T_p is the number of planning periods, N_p indicates the duration of each planning period, and N_y corresponds to the total planning horizon in years. The present value factor $PF(\tau)$, used to compute the discounted value of equal annual cash flows for investment purposes, is defined as:

$$PF(\tau) = \frac{(1+\gamma)^\tau \gamma}{(1+\gamma)^\tau - 1} \quad (22)$$

where $n_{i,\tau}^{PV}$ represents the number of PV units at node i in planning year τ . S_0^{PV} denotes the rated installation capacity of a single PV unit, while $S_{i,max}^{PV}$ signifies the maximum planned capacity of PV units at node i . Φ_{PV} is the set of candidate nodes for PV installation. $n_{i,\tau+1}^{WT} - n_{i,\tau}^{WT}$ denotes the newly planned increment in WTs at node i during the $\tau+1$ th planning cycle. A similar logic applies to other investment projects. The before-planning stage conditions are defined as:

$$n_{i,0}^{PV} = n_{i,0}^{WT} = n_{i,0}^{ESS} = n_{i,0}^{EVCS} = n_{ij,0}^{line} = 0. \quad (23)$$

Lower-level objective function:

$$\min B^{EV} = N_d \sum_{s \in S} \rho_s \sum_{i=1}^{N_{bus}} \sum_{u=1}^{N_{EV}} \sum_{t=1}^T (\pi_t^{dch} P_{s,u,i,t,\tau}^{v2g} - \pi_t^{ch} P_{s,u,i,t,\tau}^{g2v}) \quad (24)$$

s.t. (25)–(31).

B. Aggregated EVs Feasible Region

Modeling individual EVs is computationally intractable at scale. Therefore, EV fleets are aggregated into a VPP representation that preserves system-level flexibility while providing a tractable boundary for planning optimization.

$$\Gamma_i = \bigoplus_u \Gamma_{u,i} = \bigoplus_u \{W_{u,i} \mathbf{P}_{u,i} \leq M_{u,i}\} \quad (25)$$

$$\mathbf{P}_{u,i} = \left[\left[\eta_{u,i}^{ch} P_{u,i}^{g2v} \right]_{T \times 1}; \left[\frac{P_{u,i}^{v2g}}{\eta_{u,i}^{dch}} \right]_{T \times 1} \right] \quad (26)$$

$$P_{u,i}^{g2v} = \left[P_{u,i,t}^{g2v} \right]_{T \times 1} \quad P_{u,i}^{v2g} = \left[P_{u,i,t}^{v2g} \right]_{T \times 1} \quad (27)$$

$$\bar{P}_{u,i}^{g2v} = \left[\bar{P}_{u,i,t}^{g2v} \right]_{T \times 1} \quad \underline{P}_{u,i}^{g2v} = \left[\underline{P}_{u,i,t}^{g2v} \right]_{T \times 1} \quad (28)$$

$$\bar{P}_{u,i}^{v2g} = \left[\bar{P}_{u,i,t}^{v2g} \right]_{T \times 1} \quad \underline{P}_{u,i}^{v2g} = \left[\underline{P}_{u,i,t}^{v2g} \right]_{T \times 1} \quad (29)$$

$$\bar{E}_{u,i} = \left[\bar{E}_{u,i,t} \right]_{T \times 1} \quad \underline{E}_{u,i} = \left[\underline{E}_{u,i,t} \right]_{T \times 1} \quad (30)$$

$$W_{u,i} = \begin{bmatrix} -I & 0 \\ I & 0 \\ 0 & -I \\ 0 & I \\ -A & A \\ A & -A \end{bmatrix} M_{u,i} = \begin{bmatrix} -\eta_{u,i} P_{u,i}^{g2v} \\ \eta_{u,i} \bar{P}_{u,i}^{g2v} \\ -P_{u,i}^{v2g} / \eta_{u,i} \\ \bar{P}_{u,i}^{v2g} / \eta_{u,i} \\ lE_{u,i,t_a} - \bar{E}_{u,i} \\ \bar{E}_{u,i} - lE_{u,i,t_a} \end{bmatrix}. \quad (31)$$

where I is the T -dimensional identity matrix. $A = \text{diag}[\Delta T]_{T \times T}$. T is the total time horizons. EVCSs aggregate EV charging and discharging capabilities, enabling coordinated flexibility for ADN operation.

C. Distributed Energy Resources Planning and Operation Constraints

Constraints on the upper and lower limits of the installed capacity at candidate node i :

$$\begin{aligned} 0 \leq n_{i,\tau}^{PV} S_0^{PV} \leq S_{i,\tau,\max}^{PV}, 0 \leq n_{i,\tau}^{WT} S_0^{WT} \leq S_{i,\tau,\max}^{WT}, \\ 0 \leq n_{i,\tau}^{ESS} S_0^{ESS} \leq S_{i,\tau,\max}^{ESS}, 0 \leq n_{ij,\tau}^{line} S_0^{line} \leq S_{ij,\tau,\max}^{line}. \end{aligned} \quad (32)$$

Operational constraints for PV and WT systems:

$$0 \leq \tilde{P}_{s,i,t}^{WT} \leq n_i^{WT} S_0^{WT} P_{s,i,t}^{WT}, 0 \leq \tilde{P}_{s,i,t}^{PV} \leq n_i^{PV} S_0^{PV} P_{s,i,t}^{PV}. \quad (33)$$

DG operational constraints:

$$0 \leq P_{s,i,t}^{DG} \leq P_{i,\max}^{DG}, 0 \leq Q_{s,i,t}^{DG} \leq Q_{i,\max}^{DG}. \quad (34)$$

Wind and PV power curtailment formulation:

$$P_{s,i,t}^{Wta} = n_i^{WT} S_0^{WT} P_{s,i,t}^{WT} - \tilde{P}_{s,i,t}^{WT}, P_{s,i,t}^{Pva} = n_i^{PV} S_0^{PV} P_{s,i,t}^{PV} - \tilde{P}_{s,i,t}^{PV}. \quad (35)$$

ESS operational constraints:

$$\begin{cases} 0 \leq P_{s,i,t}^{ESS, ch} \leq P_{i,\max}^{ESS, ch}, 0 \leq P_{s,i,t}^{ESS, dch} \leq P_{i,\max}^{ESS, dch} \\ P_{s,i,t}^{ESS, dch} P_{s,i,t}^{ESS, ch} = 0, E_{\min}^{ESS} \leq E_{s,i,t}^{ESS} \leq E_{\max}^{ESS} \\ E_{s,i,t}^{ESS} = E_{s,i,t-1}^{ESS} + (P_{s,i,t}^{ESS, ch} \eta_{u,i}^{ch} - P_{s,i,t}^{ESS, dch} / \eta_{u,i}^{dch}) \Delta. \end{cases} \quad (36)$$

Constraints on EVCS planning:

$$N_{\min} \leq n_i^{EVCS} \leq N_{\max} \quad (37)$$

$$\sum_u z_{t,u} \leq n_i^{EVCS}. \quad (38)$$

where $z_{t,u}$ indicates whether user u is charging at time t . (38) ensures the number of charging piles smaller than EVs, which is also modeled in (2), (3).

Charging station constraints:

$$0 \leq P_{s,u,i,t}^{g2v} \leq z_{t,u,i} P^{ch} \quad (39)$$

$$P_{s,i,t}^{g2v} = \sum_u P_{s,u,i,t}^{g2v} \quad (40)$$

where $z_{t,u,i}$ is a binary variable, representing if the u -th EV is participating in charging at node i at time t .

D. Distribution Network Operation Constraints

Branch power constraints:

$$P_{i,t} = \sum_{j=1, j \neq i}^{N_B} \left(\frac{k_{ij-2}}{x_{ij}} (\theta_{i,t} - \theta_{j,t}) + \frac{k_{ij-1}}{x_{ij}} (V_{i,t} - V_{j,t}) \right) \quad (41)$$

$$Q_{i,t} = \sum_{j=1, j \neq i}^{N_B} \left(-\frac{k_{ij-1}}{x_{ij}} (\theta_{i,t} - \theta_{j,t}) + \frac{k_{ij-2}}{x_{ij}} (V_{i,t} - V_{j,t}) \right) \quad (42)$$

$$P_{ij,t} = \frac{k_{ij-2}}{x_{ij}} (\theta_{i,t} - \theta_{j,t}) + \frac{k_{ij-1}}{x_{ij}} (V_{i,t} - V_{j,t}) \quad (43)$$

$$Q_{ij,t} = -\frac{k_{ij-1}}{x_{ij}} (\theta_{i,t} - \theta_{j,t}) + \frac{k_{ij-2}}{x_{ij}} (V_{i,t} - V_{j,t}) \quad (44)$$

where $k_{ij-1} = r_{ij} x_{ij} / r_{ij}^2 + x_{ij}^2$, $k_{ij-2} = x_{ij}^2 / r_{ij}^2 + x_{ij}^2$, $P_{i,t}$, $Q_{i,t}$ are per-unit injected active and reactive power, respectively; $\theta_{i,t}$, $V_{i,t}$ are per-unit node voltage phase angle and magnitude, respectively.

Node injection power balance constraints:

$$P_{i,t} = P_{i,t}^{DN} + \tilde{P}_{i,t}^{RES} + P_{i,t}^{EVCS} + P_{i,t}^{DG} + P_{i,t}^{ESS, dch/ch} - P_{i,t}^{load} \quad (45)$$

$$Q_{i,t} = Q_{i,t}^{DN} + Q_{i,t}^{DG} - Q_{i,t}^{load}. \quad (46)$$

Node voltage constraints:

$$V_i^{\min} \leq V_{i,t} \leq V_i^{\max}. \quad (47)$$

Line power flow constraints:

$$P_{ij}^{\min} \leq P_{ij,t} \leq P_{ij}^{\max} + n_{ij}^{line} S_0^{line}. \quad (48)$$

Power purchase constraints from the upper-grid:

$$P_{s,i,t}^{DN, \min} \leq P_{s,i,t}^{DN} \leq P_{s,i,t}^{DN, \max}. \quad (49)$$

To account for the uncertainty of RES in reserve scheduling and its impact on transmission constraints and voltage profiles, the DRJCC is proposed, where the total error term is endogenously coupled with the final planning variables, rendering it a decision-dependent quantity:

$$\inf_{\Theta \in \mathbb{P}} \Pr \left\{ \begin{aligned} -\alpha_{s,i,t} \Omega_{s,t}(\mathbf{n}) &\leq r_{i,t}^+ \\ -\alpha_{s,i,t} \Omega_{s,t}(\mathbf{n}) &\geq -r_{i,t}^+ \end{aligned} \right\} \geq 1 - \varepsilon_r, \forall i. \quad (50)$$

$$\inf_{\Theta \in \mathbb{P}} \Pr \left\{ \begin{aligned} \Phi_{(\cdot, ij)}(\tilde{g} - \tilde{d}) &\leq f_{ij}^{\max} \\ \Phi_{(\cdot, ij)}(\tilde{g} - \tilde{d}) &\geq -f_{ij}^{\max} \end{aligned} \right\} \geq 1 - \varepsilon_l, \forall ij \quad (51)$$

$$\inf_{\Theta \in \mathbb{P}} \Pr \left\{ \underline{V} \leq V_{i,t} + PT_V \xi_i \leq \bar{V}, \forall i \in N_B \right\} \geq 1 - \varepsilon^V, \forall t \quad (52)$$

where the shift factor-based DC power flow model $\Phi_{(\cdot, ij)}(\cdot)$ is used. $\tilde{g}$ represents the power output like generation or ESS discharging vectors. $\tilde{d}$ represents the all kinds of loads vectors. utilizing the moment information of ambiguity sets to represent the probability distribution of uncertain variables, the aforementioned DRJCC can be reformulated using the Bonferroni approximation as follows:

$$\sqrt{\frac{1-\varepsilon_r}{\varepsilon_r}} \sqrt{a_i(\mathbf{n})^T \Sigma a_i(\mathbf{n})} \leq b_i(\mathbf{n}) - a_i(\mathbf{n})^T \mu, \forall i \quad (53)$$

For (50), the constraint parameters are defined as:

$$a_i(\mathbf{x}) = \alpha_{s,i,t} n_i, b_i(\mathbf{x}) = r_{i,t}^+ \quad (54)$$

In (51), the parameters become:

$$b_{ij}(\mathbf{x}) = f_{ij}^{\max} - \Phi_{(\cdot, ij)}(\tilde{g} - \tilde{d}). \quad (55)$$

Using the upper-bound constraint as an illustrative case: $a_i(\mathbf{x}) = \Phi_{(\cdot, ij)}(1 - \alpha_{s,i,t} n_i)$. The corresponding parameters for the lower-bound constraint can be derived analogously to obtain the appropriate $a_i(\mathbf{x})$, $b_i(\mathbf{x})$.

IV. BI-LEVEL OPTIMIZATION BETWEEN DSO AND EVCS

The original formulation is a bilevel mixed-integer nonlinear program. After linearizing and replacing the lower level with Karush-Kuhn-Tucker (KKT) conditions, it is reformulated as single-level MILP solvable by commercial solvers.

A. KKT Conditions and Linearization

Based on the KKT optimality condition, the Lagrange function of the lower-level problem in (24) can be written as:

$$\begin{aligned} L(X) = & N_d \sum_{s \in S} \rho_s \sum_{i=1}^{N_{bus}} \sum_{u=1}^{N_{EV}} \sum_{t=1}^T [\pi_t^{ch} P_{s,u,i,t}^{g2v} - \pi_t^{dch} P_{s,u,i,t}^{v2g} - \lambda_{1,u,i,t} P_{s,u,i,t}^{g2v} \\ & + \bar{\lambda}_{1,u,i,t} (P_{s,u,i,t}^{g2v} - \bar{P}_{s,u,i,t}^{g2v}) - \lambda_{2,u,i,t} P_{s,u,i,t}^{v2g} + \bar{\lambda}_{2,u,i,t} (P_{s,u,i,t}^{v2g} - \bar{P}_{s,u,i,t}^{v2g}) \\ & + (-\eta_{u,i,t}^{ch} P_{s,u,i,t}^{g2v} \Delta t + P_{s,u,i,t}^{v2g} \Delta t / \eta_{u,i,t}^{dch} + \bar{E}_{s,u,i,t+1} - E_{s,u,i,t_{arr}}) \lambda_{3,u,i,t} \\ & + (-\eta_{u,i,t}^{ch} P_{s,u,i,t}^{g2v} \Delta t - P_{s,u,i,t}^{v2g} \Delta t / \eta_{u,i,t}^{dch} - \bar{E}_{s,u,i,t+1} + E_{s,u,i,t_{arr}}) \bar{\lambda}_{3,u,i,t}]. \end{aligned} \quad (56)$$

The first order derivatives of the Lagrange function with respect to the decision variables of the lower-level problem can be derived as follows stationary conditions:

$$\begin{aligned} -\pi_t^{dch} - \lambda_{2,u,i,t} + \bar{\lambda}_{2,u,i,t} + \lambda_{3,u,i,t} \Delta t / \eta_{u,i,t}^{dch} - \bar{\lambda}_{3,u,i,t} \Delta t / \eta_{u,i,t}^{dch} &= 0 \\ \pi_t^{ch} - \lambda_{1,u,i,t} + \bar{\lambda}_{1,u,i,t} - \eta_{u,i,t}^{ch} \lambda_{3,u,i,t} \Delta t + \eta_{u,i,t}^{ch} \bar{\lambda}_{3,u,i,t} \Delta t &= 0. \end{aligned} \quad (57)$$

Complementary slackness constraints:

$$\begin{aligned} \lambda_{1,u,i,t} \perp P_{s,u,i,t}^{g2v}, \bar{\lambda}_{1,u,i,t} \perp (P_{s,u,i,t}^{g2v} - \bar{P}_{s,u,i,t}^{g2v}) \\ \lambda_{2,u,i,t} \perp P_{s,u,i,t}^{v2g}, \bar{\lambda}_{2,u,i,t} \perp (P_{s,u,i,t}^{v2g} - \bar{P}_{s,u,i,t}^{v2g}) \end{aligned} \quad (58)$$

$$(-\eta_{u,i,t}^{ch} P_{s,u,i,t}^{g2v} \Delta t + P_{s,u,i,t}^{v2g} \Delta t / \eta_{u,i,t}^{dch} + \bar{E}_{s,u,i,t+1} - E_{s,u,i,t_{arr}}) \perp \lambda_{3,u,i,t} \quad (59)$$

$$(\eta_{u,i,t}^{ch} P_{s,u,i,t}^{g2v} \Delta t - P_{s,u,i,t}^{v2g} \Delta t / \eta_{u,i,t}^{dch} - \bar{E}_{s,u,i,t+1} + E_{s,u,i,t_{arr}}) \perp \bar{\lambda}_{3,u,i,t} \quad (60)$$

$$\lambda_{1,u,i,t}, \bar{\lambda}_{1,u,i,t}, \lambda_{2,u,i,t}, \bar{\lambda}_{2,u,i,t}, \lambda_{3,u,i,t}, \bar{\lambda}_{3,u,i,t} \geq 0, \forall u, i, t. \quad (61)$$

The bilevel problem is need to be reformulated to single-level via KKT conditions, where the lower-level model should become a convex linear program after linearizing bilinear terms using standard big-M techniques [33]. For example, formula (8) can be transformed into:

$$P_{s,u,i,t}^{g2v} \leq z_{s,u,i,t}^{g2v} \cdot \bar{P}_{s,u,i,t}^{g2v}, P_{s,u,i,t}^{v2g} \leq (1 - z_{s,u,i,t}^{v2g}) \cdot \bar{P}_{s,u,i,t}^{v2g} \quad (62)$$

Formula (59) can be transformed into:

$$\lambda_{3,u,i,t} \leq z_{3,u,i,t} \cdot M \quad (63)$$

$$\begin{pmatrix} -\eta_{u,i,t}^{ch} P_{s,u,i,t}^{g2v} \Delta t + P_{s,u,i,t}^{v2g} \Delta t / \eta_{u,i,t}^{dch} \\ + \bar{E}_{s,u,i,t+1} - E_{s,u,i,t_{arr}} \end{pmatrix} \leq (1 - z_{3,u,i,t}) \cdot M \quad (64)$$

where M is a large number, $z_{3,u,i,t}$ is a binary variable. Therefore, formula (57), (58) and (60) can be similarly linearized to improve computational efficiency.

After fixing admission decisions and linearizing the feasible-region constraints, the lower-level EV aggregation model becomes a continuous linear program, satisfying the convexity and strong-duality conditions required for KKT reformulation.

B. Single-Level Optimization Transition

By employing the KKT optimality conditions, the bilevel programming problem can be transformed into the following single-level problem:

$$\begin{aligned} \min C_1 = & C^{inv} + C^{line} + C^{ope} + C^{aban} + C^{DG} + \\ & T_y \sum_{s \in S} \rho_s \sum_{i=1}^{N_{bus}} \sum_{u=1}^{N_{EV}} \sum_{t=1}^T \begin{pmatrix} -\lambda_{1,u,i,t} \bar{P}_{s,u,i,t}^{g2v} + \bar{\lambda}_{2,u,i,t} \bar{P}_{s,u,i,t}^{v2g} \\ -\lambda_{3,u,i,t} (E_{s,u,i,t+1} - E_{s,u,i,t_{arr}}) \\ + \bar{\lambda}_{3,u,i,t} (\bar{E}_{s,u,i,t+1} - E_{s,u,i,t_{arr}}) \end{pmatrix} \end{aligned} \quad (65)$$

s.t. (5)–(13), (25)–(55), (57)–(64).

By introducing MILP, the model simultaneously determines investment decisions in the upper level and validates operational feasibility in the lower level, which can be solved by GUROBI in MATLAB. The overall framework of the proposed method is shown in Fig. 1:

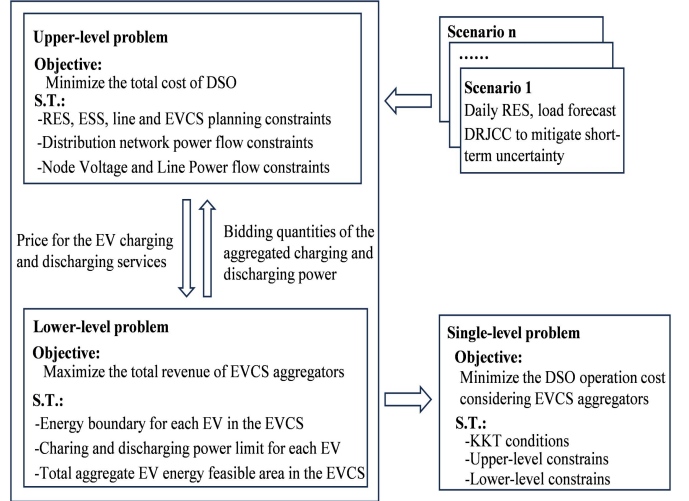


Fig. 1 General framework of the proposed bilevel planning approach.

V. SIMULATION AND RESULT ANALYSIS

In this section, a series of simulation tests on different scale of distribution networks (25-node, IEEE 141-node) planning for large scale of EVs accommodation are employed to validate the proposed approach using GUROBI in MATLAB 2024a. All MILP models were solved by GUROBI with a relative optimality gap of 0.05%, which provides near-optimal solutions while maintaining computational tractability for large-scale planning cases. All tests have been conducted on a PC with an Intel Core i7 CPU at 2.60 GHz and 16 GB of RAM.

A. 25-Node Distribution Network Planning

To demonstrate the effectiveness and economic benefits of the proposed bi-level stochastic ADN planning model under uncertainty, the proposed approach is conducted on the 25-node ADN planning as shown in Fig. 2, which corresponds to an actual medium-voltage distribution feeder located in a district of Zhongshan, Guangdong Province, China. The planning horizon is 15 years. Data were collected at 30-minute intervals. Parameter γ in (22) is set to 0.05, and the other model

parameters are shown in Table II. Topological parameters like line resistance, line reactance, etc. can be accessed from [39].

The specific cost parameters presented in Table II are primarily derived from references [4], [12], and [23], and are further supplemented by guidelines and installation demand recommendations from China Southern Power Grid Co [34].

TABLE II
MODEL PARAMETER

Item	Data
PV Capacity Investment	571.43k\$/MW
PV Rated Capacity	0.01MW
WT Capacity Investment	857.14k\$/MW
WT Rated Capacity	0.01MW
ESS Capacity Investment	214.29k\$/MW
ESS Rated Capacity	0.01MW
PV Unit Electricity O&M Cost	4.29\$/MW
WT Unit Electricity O&M Cost	7.14\$/MW
Curtailment Cost of WT and PV	14.29\$/MW
EVCS Investment Cost	5000\$/unit
EVCS Maintenance Cost	500\$/unit/year
EVCS rated Charging Power	0.08MW
Line Expansion Cost	700k\$/MW
Total Number of EVs	200
Typical Days	4
Converge Gap for 25 Nodes	0.01%
Converge Gap for IEEE 141/300	0.05%

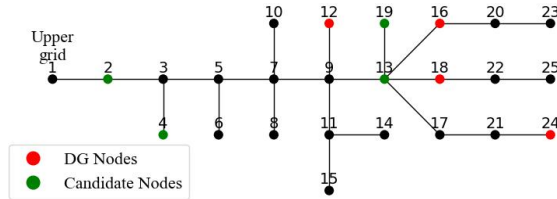


Fig. 2 Pre-Planning Topology in 25-Node Test Feeder

However, the pre-planned distribution network may lack sufficient RES. Additionally, voltage stability is poor, especially during peak load periods and the number of charging piles is limited, unable to meet the charging demands of EVs. Consequently, typical days following scenario reduction are shown below to lay a foundation for implementing proposed

planning method. To enhance the accuracy of clustering results, A total of nine statistical metrics is considered, including: mean, variance, total sum, maximum/minimum, skewness, kurtosis, root mean square (RMS), coefficient of variation, and autocorrelation coefficient [11], [26].

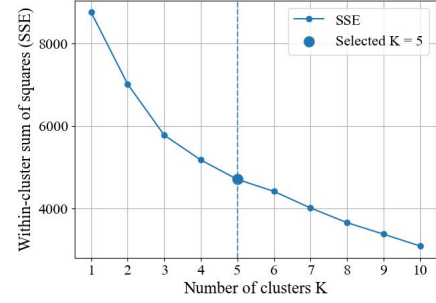


Fig. 3 Elbow Method for Determining the Number of Typical Days.

TABLE III
CLUSTERING RESULTS

Cluster	Number of Days	Percentage
1	109	29.9%
2	132	36.2%
3	69	18.9%
4	2	0.5%
5	53	14.5%

In Fig. 3, the number of typical days was determined using the elbow method. K-means clustering was performed with the number of clusters varying from 1 to 10, and the within-cluster sum of squared errors was calculated for each case. The SSE decreases rapidly when K increases from 1 to 4, while the marginal reduction becomes much smaller after K=5. Therefore, K=5 was selected to balance scenario representativeness and computational tractability.

However, cluster 4 contained only 2 days, corresponding to 0.5% of the annual samples. This cluster was treated as a low-frequency abnormal scenario, not retained as an independent typical day in the model. The probabilities of the four retained typical days are normalized according to the number of days in each retained cluster. i.e., $\rho_k = N_k / \sum_k N_k$.

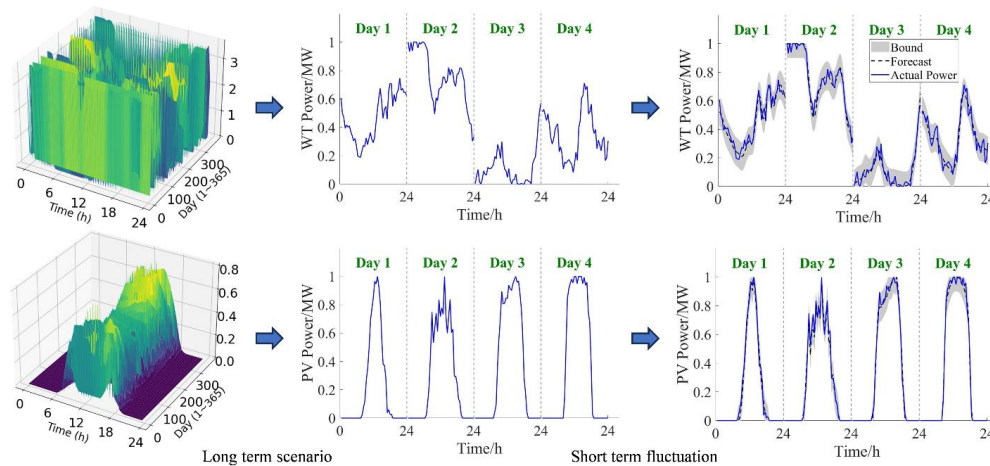


Fig. 4 K-means Scenario Reduction of Renewable Generation.

Fig. 4 illustrates the K-means-based scenario reduction process for WT and PV generation. The raw annual profiles are reduced into representative daily scenarios, which retain the

main temporal characteristics of WT variability and PV daily patterns. The comparison between forecasted and actual outputs

further shows that the reduced scenarios preserve intra-day uncertainty while reducing computational burden.

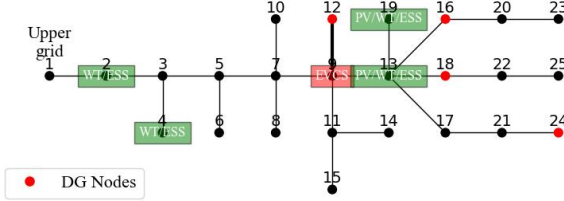


Fig. 5 Post-Planning Topology in 25-Node Test Feeder

Case 1: DSO planning with regular EV charging but no RES.

Case 2: DSO planning with RES and regular EV charging.

Case 3: DSO planning in multi-scenario RES with DD-DRJCC ($\varepsilon_r = 0.1$) and regular EV charging.

Case 4: DSO planning in multi-scenario RES with normal-DRJCC ($\varepsilon_r = 0.1$) and coordinated EV aggregate charging.

Case 5: DSO planning in multi-scenario RES with DD-DRJCC ($\varepsilon_r = 0.1$) and coordinated EV aggregate charging.

According to both the literature [35]-[36] and prevailing regulations [37], the violation probability ε is typically set at 5% under day-ahead scheduling, while an initial planning stage may adopt a more lenient value such as 10% to account for preliminary uncertainties. During intraday scheduling, this value can be refined to around 1%. If ε is set overly conservatively, the required reserve capacity and associated costs will increase substantially, adversely affecting the initial planning decisions. Conversely, an excessively high ε may compromise system reliability and safety.

TABLE IV

COMPARISON OF STRATEGIES FOR SYSTEM PLANNING IN 25-NODE

Case	PV	WT	ESS(MW)
Case 2	2.5(2)	2.46(2)	1.5(2), 1.5(4), 1.5(13), 1.5(19)
Case 3	2.5(13), 1.56(19)	2.47(2), 2.45(4), 0.24(13), 0.01(19)	1.49(2), 1.5(4), 1.5(13), 1.5(19)
Case 4	1.65(2), 2.5(13)	2.5(2), 0.29(4), 2.17(13)	1.47(2), 0.11(4), 1.25(19)
Case 5	2.5(13), 1.7(19)	2.30(2), 2.42(4), 0.01(13), 0.01(19)	0.03(2), 1.43(4), 0.52(13)

Candidate RES and ESS planning nodes are 2, 4, 13, and 19. The post-planned ADN is shown in Table IV. Strategic sizing and siting of distributed resources can effectively improve

voltage stability and power balance, while also providing flexible power support for EVCS in node 9.

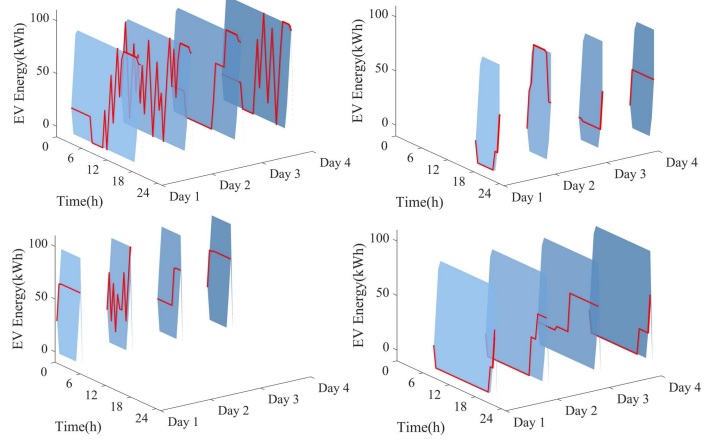


Fig. 6 Multi-Dimensional Energy Feasible area of single EV

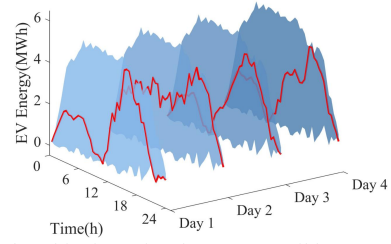


Fig. 7 Aggregated Multi-Dimensional Energy Feasible area of EVs

Fig. 6 presents 4 EVs' energy feasible regions and actual energy consumption curve. Notably, the actual energy curves consistently remain within the feasible regions, and EVs exhibit distinct charging patterns.

The aggregated EVs operation is in Fig. 7, reducing peak demand while maintaining total energy within the collective feasible regions. The actual aggregated energy closely tracks the lower boundary of the feasible regions during off-peak hours (00:00-06:00), but approaches the upper boundary (5 MWh) during evening peaks (18:00-22:00), suggesting strategic load shifting, confirming the effectiveness of fleet-level energy management in balancing grid constraints with EV user needs.

TABLE V

COMPARISON OF STRATEGIES FOR PLANNING COST IN 25-NODE

Case	Investment(k\$)	Line(k\$)	DN(k\$)	O&M(k\$)	Aban.(k\$)	DG(k\$)	Reserve(k\$)	Total(k\$)	Time(s)
Case 1	31.793	7.707	1093.957	-	-	2743.686	-	3877.144	-
Case 2	622.509	7.707	528.100	155.982	0.698	1189.073	-	2504.070	18
Case 3	805.903	3.854	35.076	279.803	155.431	932.096	58.560	2270.723	32
Case 4	728.279	3.854	34.408	288.543	120.057	891.221	157.182	2223.543	538
Case 5	695.316	3.854	34.350	281.899	112.662	920.352	52.604	2101.037	409

TABLE VI

SENSITIVITY ANALYSIS OF PLANNING SYSTEM UNDER DIFFERENT NUMBERS OF EVs IN 25-NODE

EV demand (percentage*base)	ESS(k\$)	EVCS Charging piles (unit)	Annual Wind Curtailment (MWh)	Total Cost (k\$)	Computational Time (s)
50%	96.618	15	104.238	2123.207	60
100% (Base)	67.921	30	97.595	2116.355	276
200%	42.528	60	86.277	2101.037	409
300%	30.554	90	74.050	2088.214	1023

TABLE VII
COMPARISON OF STRATEGIES FOR PLANNING COST IN IEEE-141 AND IEEE-300

Case	Investment(k\$)	Line(k\$)	DN(k\$)	O&M(k\$)	Aban.(k\$)	DG(k\$)	Reserve(k\$)	Total(k\$)	Times(s)
Case 6	3379.825	67.440	1789.947	835.708	6.860	6784.775	-	12864.555	164
Case 7	4316.685	27.939	30.830	1534.921	108.807	3897.622	238.901	10155.705	986
Case 8	17684.452	20.232	724.313	7027.750	473.259	31318.411	1311.244	58559.661	2153

TABLE VIII
EFFECT OF DIFFERENT EV CAPACITIES ON PLANNING COST IN 25-NODE

Average Capacity	Investment(k\$)	Line(k\$)	DN(k\$)	O&M(k\$)	Aban.(k\$)	DG(k\$)	Reserve(k\$)	Total(k\$)
60kWh	677.355	3.854	34.484	272.348	110.103	938.074	51.176	2087.393
70kWh	687.884	3.854	34.157	275.625	110.331	926.879	51.620	2090.350
80kWh	691.256	3.854	34.174	278.040	111.846	922.342	52.063	2093.575
90kWh	694.834	3.854	34.233	280.634	111.966	918.793	52.420	2096.734
100kWh	695.316	3.854	34.350	281.899	112.662	920.352	52.604	2101.037

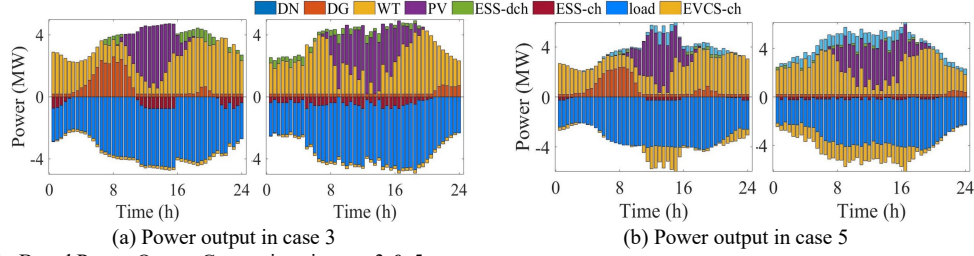


Fig. 8 Multi-RES Scenario-Based Power Output Comparison in case 3 & 5.

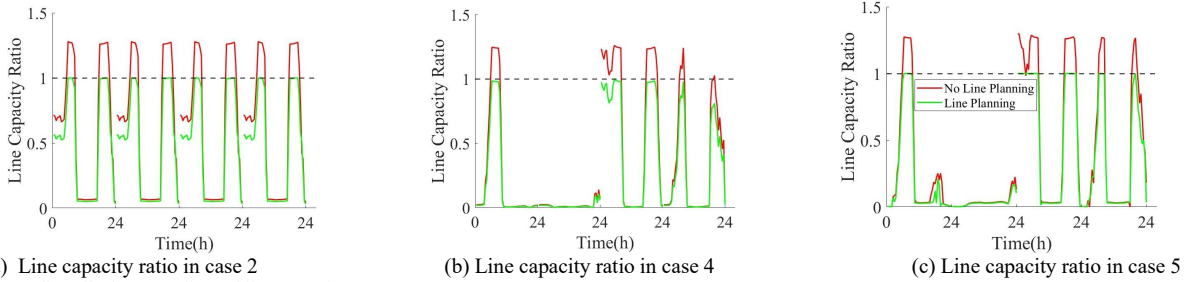


Fig. 9 The power flow limit-exceeding of line 9-12 in case 2, 4, 5.

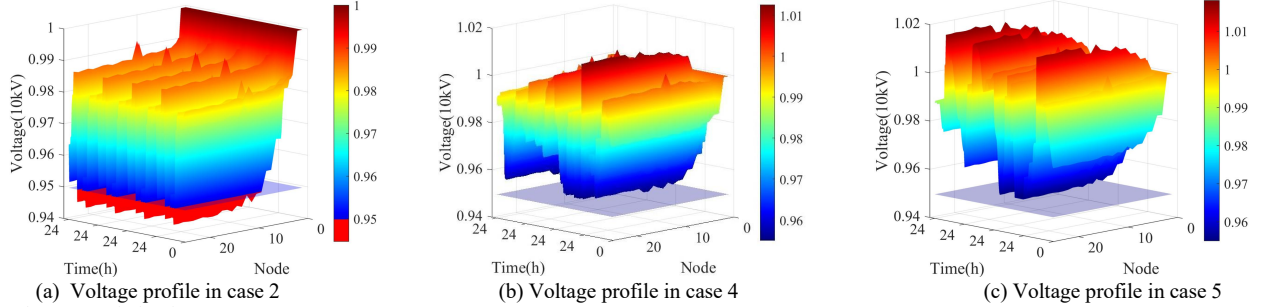


Fig. 10 Voltage Regulation in case 2, 4, 5.

In Table V, comparative analysis of cases reveals significant variations driven by different strategies. Case 1 incurs the highest total cost, primarily from DG cost and DN cost. Case 2 introduces RES but maintains conservative operation, resulting in lower total costs yet inefficient resource allocation, evidenced by high DG cost 1189.073k\$ and DN cost 528.1k\$. Case 3's DRJCC optimization reduces DG cost but retains substantial ESS investments with 1.5MW across nodes, indicating suboptimal utilization of flexible resources. Notably, Case 5 achieves the lowest total cost through coordinated EV aggregation, reducing DN costs to 34k\$ and ESS capacity to 0.03–1.43 MW by leveraging idle EVs for virtual storage. These demonstrate that EVs integration (Case 5) optimally balances cost reduction with resilience, overcoming the limitations of

conservative planning and operation (Case 2) and over-reliance on physical storage (Case 3).

Fig. 8 compares the power output (2 of 4 typical days) of Case 3 and Case 5 in the 25-node system. The left panel demonstrates how Case 3's RES allocation leads to significant curtailment during peak generation hours (e.g., 12:00-15:00), while Case 5's EV-VPP coordination enables fuller utilization of renewable energy, reducing curtailment by 27.6%. The right panel reveals that Case 5's WT output (2.3-2.42 MW) better matches temporal demand patterns than Case 3's uniform deployment (2.47 MW), validating the economic benefits of scenario-aware planning.

Fig. 9 examines line congestion mitigation, where Case 2 (baseline RES planning) exhibits severe power flow violations

on Line 9-12 (exceeding 1.8p.u. during 18:00-20:00). In contrast, Case 5's EV-VPP coordination maintains flows within 0.95p.u. limits through bidirectional power regulation, reducing peak congestion by 47.3%. This underscores how aggregated EVs act as flexible resources to alleviate branch overloads.

Fig. 10 demonstrates voltage regulation improvements, with Case 5 achieving tighter voltage bands (0.955-1.02 p.u.) than that of Case 2 (0.94-1.0 p.u.). The nodal voltage deviation times decreases from 4.2% to 1.8%, proving EV reactive power support enhances voltage stability under high RES penetration.

The results, as depicted in Fig. 9 and 10, demonstrate that both methods in Case 4 and Case 5 maintain operational safety and stability within acceptable limits. Although the normal DRJCC slightly outperforms in terms of voltage regulation and power flow limit adherence, Table V reveals that the DD-DRJCC approach achieves a 5.5% reduction in planning costs. This comparative analysis validates our model and suggests that the normal DRJCC approach, while slightly more conservative, offers a marginally better performance in terms of system reliability at a higher cost. The trade-off between cost and reliability is thus evident, highlighting the importance of selecting the appropriate planning strategy based on specific operational requirements and constraints.

B. Sensitivity Analysis in 25-Node System

This section conducts sensitivity analysis to evaluate the impact of varying EV fleet sizes and EV charging piles on key planning metrics, while all other parameters of the 25-node distribution network are held invariant.

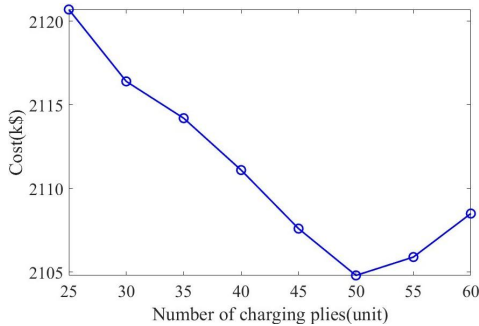


Fig. 11 Cost under different EV charging piles numbers.

Table VI demonstrates relationship between EV fleet size and system costs: Increasing the fleet from 50% to 300%, EVs reduces ESS costs from 96.6k\$ to 30.6k\$. Due to the larger fleet provides more storage capacity, reducing reliance on ESS. The number of required charging piles scales linearly (15→90), but the per-EV infrastructure cost decreases due to improved utilization efficiency. Larger fleets absorb more excess renewable generation, reducing curtailment from 104.2 MW to

74.1 MW, thus enhancing grid sustainability.

Fig. 11 shows the total system costs under different number of EV charging piles with 100 EVs. When the number of piles smaller than 50, the cost decrease as the increasing of piles, because of more idle EV resources can be used to participate in VPP services. When the number is larger than 50, the cost increases as the increasing of piles, due to the investment of EVCS may not offset the revenue of DSO, or even lead to additional system cost like line reinforcement.

To assess the robustness of the planning methodology, the following discussion examines the influence of EV battery capacities on planning outcomes. This analysis aims to provide insights into how varying EV capacities affect the overall planning costs and resource allocation strategies.

Table VIII illustrates the impact of varying EV capacities on planning costs within a 25-node network. It is evident that as the capacity of EVs increases, the total planning cost also rises. This increase is primarily attributed to the load characteristics of EVs; as the load capacity expands, the demand for electricity generation correspondingly increases, thereby escalating the overall cost. Additionally, the investment cost continues to rise, which implies an expansion in the capacity of internal renewable resources such as wind and solar power. Interestingly, despite the increase in capacity, the cost associated with DG exhibits a downward trend. This can be attributed to the enhanced ability of larger EV capacities to absorb and utilize renewable energy resources more effectively, thereby reducing the relative cost of DG. This observation underscores the potential for EVs to play a significant role in integrating and optimizing the use of renewable energy within electrical networks, contributing to more cost-effective and sustainable energy planning strategies.

C. IEEE-141 and IEEE-300 Distribution Network Planning

To further validate the scalability and robustness of the proposed bi-level stochastic planning model, a second case study is conducted on a large-scale IEEE 141-node distribution system. This scenario evaluates the model's performance under increased network complexity, higher node density, and greater variability in renewable resource and EV penetration. Two distinct planning strategies are compared.

Case 6: Deterministic optimization with regular EV charging in IEEE-141.

Case 7: Multi-scenario stochastic optimization with DD-DRJCC ($\epsilon_r=0.1$) with coordinated EV charging in IEEE-141.

Case 8: Multi-scenario stochastic optimization with DD-DRJCC ($\epsilon_r=0.1$) with coordinated EV charging in IEEE-300.

TABLE IX
COMPARISON OF STRATEGIES FOR SYSTEM PLANNING IN IEEE-141

Case		Planning Capacity/MW(Position)
PV	Case 6	3.87(30), 6.46(42), 4.64(65), 4.17(97), 2.54(104), 5.64(122)
	Case 7	2.81(30), 6.64(42), 4(65), 1.5(97), 3.6(104), 4.78(122)
WT	Case 6	0.53(30), 2.11(42), 4.42(65), 1.19(97), 4.08(128)
	Case 7	4.86(30), 7.5(42), 0.01(65), 7.5(93), 4.32(97), 0.01(104), 0.01(122), 5.97(128)
ESS	Case 6	5(30), 5(42), 5(65), 5(93), 5(97), 5(104), 5(122), 5(128)
	Case 7	5(30), 5(42), 5(65), 1.46(97), 3.16(104), 5(122)

Line	Case 6	0.9(1-2), 0.7(2-3), 0.6(3-4), 0.5(4-5), 2(5-6), 1.7(6-7), 0.5(42-43), 0.1(43-44)
	Case 7	0.4(5-6), 0.6(42-43), 0.1(43-44), 0.6(7-88), 0.8(88-89), 0.2(92-93), 0.2(127-128)

Due to the page limit, results of post-planning topologies for the IEEE 141-node test system under two distinct strategies can be accessed from [39]. Table VII highlights the Case 7's cost efficiency. For Renewable Resource Allocation: In Case 6, the deterministic approach deploys uniform renewable generation capacities, with PV capacities ranging from 2.54 MW to 6.46 MW across six nodes and relatively small WT deployments with max 4.42 MW at node 65. In contrast, Case 7 exhibits a more flexible and strategically concentrated allocation. WT capacities are heavily assigned to nodes 42 and 93 with 7.5 MW each, while PV installations from 4.17 MW to 1.5 MW are downsized at nodes like 97. This adjustment reflects a scenario-aware planning strategy that aligns generation deployment with localized resource profiles and system constraints. The computational times for Case 5, Case 7, and Case 8—each employing the same optimization methodology—are reported as 409 s, 986 s, and 2153 s, respectively. These results suggest that the required computational time exhibits an approximately linear relationship with the scale of the network.

A notable difference arises in ESS deployment. Case 6 installs 8 identical 5 MW ESS units across the network (totaling 40 MW), regardless of nodal or temporal load variations. However, Case 7 demonstrates selective downsizing, with ESS capacities at nodes 97 and 104 reduced to 1.46 MW and 3.16 MW, respectively, while retaining 5 MW units at high-priority nodes. This targeted allocation reduces redundancy and leverages EVs' flexibility to replace fixed storage.

Line investment decisions show a stark contrast. Case 6 adds eight new branches (e.g., 1–2, 2–3, ..., 42–43), incurring higher infrastructure costs and indicating broader grid stress. In contrast, Case 7 introduces only six lines upgrades, with a 58.6% reduction from 67.44k\$ to 27.94k\$ in total line investment. The selection of reinforcement paths (e.g., 5–6, 7–88, 127–128) in Case 7 reflects congestion-aware planning that reinforces vulnerable radial paths without over-provisioning.

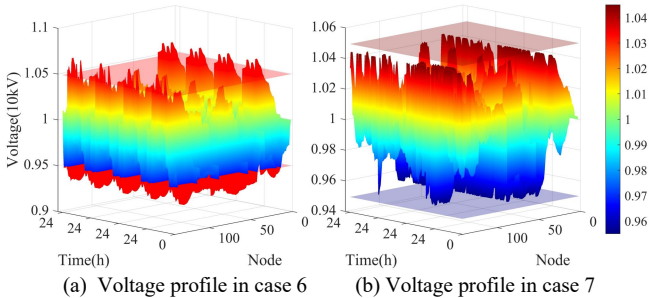


Fig. 12 Voltage Regulation in case 6 & 7.

Table VII summarizes the cost breakdown for both strategies, revealing substantial savings in Case 7. O&M Costs increase in Case 7 (+83.7%) due to the enhanced operation of RES and EV coordination. However, these are offset by capital savings in DG, ESS, and DN categories. Reserve cost (new in Case 7: 238.9k\$) reflects a small price for resilience under stochastic

uncertainty, ensuring feasibility under worst-case deviations. Overall, Case 7 reduces the total system cost by 2.71 M\$ (about 21%) compared to the deterministic baseline, showcasing the economic viability of DRJCC-based stochastic planning with EV aggregation.

Fig. 12 examines voltage profiles across all 141 nodes. Key observations include: Case 6: Voltage drops below 0.95 p.u. on part of the system nodes, indicating serious under-voltage issues in peripheral branches. Case 7: Voltages are well-regulated within [0.95p.u., 1.05p.u.]. The improvement in voltage deviation is directly attributed to two aspects, e.g., EV-VPPs providing reactive power support and charging/discharging coordination to stabilize node voltages.

VI. CONCLUSION

This paper proposes a bi-level, stochastic ADN planning framework that co-optimizes generators, storages, network reinforcement, and EV aggregated integration under long-short term uncertainties. Simulation results confirm the merits of the proposed scheme: Case 5 reduces total planning cost by up to 16.1% and 7.5% relative to Case 2 and 3 on the 25-node, and Case 7 reduces total cost by up to 21.1% compared to Case 6 on IEEE 141-node systems while simultaneously mitigating renewable curtailment, line congestion, and voltage excursions through the coordinated flexibility of EV VPPs. Additionally, Case 5 demonstrates a more significant cost reduction compared to Case 4, underscoring the enhanced efficiency of the proposed planning framework when coordinating EV flexibility with high renewable penetration. Furthermore, sensitivity analysis in this study provides critical insights into how varying EV properties impact planning costs, thereby informing more robust and cost-effective ADN planning strategies. Although the current EV model assumes immediate departure upon charge completion, practical extensions could exploit prolonged parking durations to further enlarge the operational envelope. Future work will also refine the representation of high-penetration renewable variability to support carbon-neutral planning objectives.

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